

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

BIG DATA ANALYTICS [IT-4410]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss in detail types of big data analytics taking relevant examples. (5)
 - A)
 - B) A leading e-commerce company is expanding its operations globally and dealing with an enormous amount of data generated from customer transactions, product searches, social media interactions, and sensor data from warehouses. However, they face multiple challenges, such as handling the rapid flow of data, ensuring its accuracy, managing different data formats, and extracting meaningful insights. (5)

As a data analyst, predict which key characteristics of big data are crucial in addressing these challenges and suggest strategies to manage them effectively.
- 2) An MNC is optimizing its HDFS storage settings across a large data cluster consisting of 10 DataNodes. Apply your understanding of HDFS to illustrate the purpose of setting a replication factor and how it contributes to fault tolerance. Consider the company decides to reduce the replication factor from 3 to 2 to save storage space - what trade-offs does this decision involve in terms of data availability and reliability? If the replication factor is set to 2, determine how many DataNodes will hold a copy of each data block, and describe what would happen if both nodes storing a specific block were to fail simultaneously. (5)
 - A)
 - B)

An e-commerce company named ShopZilla receives millions of customer reviews and ratings for various products. These reviews are stored as large text files in their distributed file system. The company wants to extract insights such as:

- The number of reviews per product
- The average rating per product
- Identification of products with the highest number of reviews and best ratings

ProductID	UserID	Rating	ReviewText
P1234	U001	4	Good quality and fast delivery
P1234	U012	5	Excellent product!
P5678	U003	3	Average experience

In the context of this case study, define the Map and Reduce steps needed to calculate the total number of reviews per product. Deduce the Mapper and Reducer code (in Java or Python) for calculating the average rating for each product.

- 3) You're working on a project that involves processing extremely large datasets using Spark. Determine how would you optimize your Spark job to handle data skew and ensure efficient resource utilization? Illustrate the strategies and techniques you would employ. (5)
- A)
- B) You are optimizing a Spark job that involves multiple transformations on large datasets. Apply your understanding to illustrate the difference between Narrow and Wide dependencies with suitable examples. Determine how Spark functions like map, flatMap, groupByKey, and reduceByKey relate to these dependencies and how they affect execution and performance. (5)
- 4) Determine the storage strategy followed with relevant examples of databases where ArangoDB and Cassandra applications are built upon. (5)
- A)
- B) (5)

Correlate below MongoDB commands to its equivalent SQL commands

```
a) db.students.find({ department: "CSE", age: { $gt: 20 } })
b) db.employees.update(
  { department: "HR", experience: { $gt: 5 } },
  { $set: { promotionEligible: true } },
  { multi: true }
)
c) db.orders.count({
  status: "delivered",
  totalAmount: { $gt: 500 }
})
d) db.students.find({ marks: { $gt: 80 } }).limit(3)
db.stud.remove({ class: { $nin: ["TE", "BE"] },
"marks.dmsa": { $lt: 40 } })
```

- 5) Explain how a data science team can use Spark ML Pipelines to build a sentiment analysis model from customer review data labeled as positive or negative. Describe the purpose of each stage in the pipeline, including how transformers and estimators help convert raw text into feature vectors and train the model (5)
- A)
- B) A data scientist is developing a recommendation system using Spark MLlib and needs to perform matrix-based computations to represent user-item interactions. Since the dataset is small and can fit on a single machine, the scientist is considering using local matrices for efficient computation. Based on your understanding of Spark MLlib's support for dense and sparse matrix formats, explain when it is appropriate to use each type. (5)
- Illustrate how data is stored internally in both formats, including details such as column-major order and index structures. Finally, illustrate how to create both a dense matrix and a sparse matrix using the appropriate Scala API in MLlib.

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